

Python Complete Roadmap (Beginner to Advanced)

Module 0: Computer Fundamentals (Before Python)

This module helps you understand how computers work before writing code.

1. What is a computer?

- Definition
 - Characteristics
 - Components
 - Input Devices
 - Output Devices
 - CPU
 - Memory
 - Storage
-

2. Hardware

- Definition
 - Types
 - Examples
-

3. Software

- Definition
 - Types
 - System Software
 - Application Software
 - Utility Software
-

4. Difference Between Hardware and Software

5. Operating System (OS)

- Definition
 - Why OS is needed
 - Functions of OS
 - Examples
 1. Windows
 2. Linux
 3. macOS
 4. Android
 5. iOS
-

6. Programming

- What is Programming?
 - Why Programming?
 - How Programs Work
 - Programming Cycle
-

7. Programming Languages

Generation of Languages

- 1GL
- 2GL
- 3GL
- 4GL
- 5GL

Examples

8. Types of Programming Languages

- Low-Level Language
- High-Level Language

Difference

9. Compiler

- Definition
- Working
- Advantages
- Disadvantages

10. Interpreter

- Definition
- Working
- Advantages
- Disadvantages

11. Compiler vs Interpreter

12. Static Language

Examples

- C
- C++

Characteristics

13. Dynamic Language

Examples

- Python
- JavaScript

Characteristics

14. Static vs Dynamic Language

Module 1: Introduction to Python

1. What is Python?

- Definition
 - Why Python is popular
-

2. History of Python

- Creator
 - Why it was created
 - Evolution
-

3. Why Python?

- Simplicity
 - Easy syntax
 - Large community
-

4. Features of Python

- Easy
- Readable
- Portable
- Free
- Open Source
- Interpreted
- Dynamic Typing
- Object-Oriented
- Huge Libraries
- Cross Platform

- Extensible
 - Embeddable
-

5. Advantages

6. Disadvantages

7. Applications of Python

- Web Development
 - AI
 - Machine Learning
 - Deep Learning
 - Data Science
 - Automation
 - Cyber Security
 - Networking
 - Game Development
 - IoT
 - Desktop Apps
-

8. Python Libraries

What is a Library?

Popular Libraries

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

- TensorFlow
 - PyTorch
 - OpenCV
 - Flask
 - Django
 - BeautifulSoup
 - Selenium
-

9. Python Frameworks

10. Python Versions

Python 1.x

Python 2.x

Python 3.x

Latest versions

Difference between Python 2 and Python 3

11. Installation

- Install Python
 - Install VS Code
 - Install Jupyter Notebook
 - Verify Installation
-

12. First Python Program

```
print ("Hello World")
```

13. Python Execution Process

Source Code

↓

Byte Code

↓

Python Virtual Machine (PVM)

↓

Output

14. Keywords

15. Identifiers

16. Comments

Single-line

Multi-line

17. Indentation

18. Coding Standards

PEP 8

Naming Conventions

Module 2: Variables (Your First Actual Coding Topic)

This is where most Python courses start, but you'll understand it much better after Module 0 and Module 1.

Topics:

- What is a Variable?
- Why Variables?

- Memory Allocation
 - Variable Naming Rules
 - Identifiers
 - Assignments
 - Multiple Assignments
 - Dynamic Typing
 - Variable Swapping
 - Memory Address
 - `id ()`
 - `type ()`
 - `del`
 - Input from User
 - Output Formatting
-

Module 3: Data Types

- `int`
 - `float`
 - `complex`
 - `bool`
 - `str`
 - `list`
 - `tuple`
 - `set`
 - `dictionary`
 - `None`
-

Module 4: Operators

- Arithmetic

- Assignment
 - Comparison
 - Logical
 - Bitwise
 - Membership
 - Identity
-

Module 5: Input and Output

- `input ()`
 - `print ()`
 - Formatting
 - f-string
 - `format ()`
-

Module 6: Conditional Statements

- `if`
 - `if else`
 - `elif`
 - Nested `if`
-

Module 7: Loops

- `while`
 - `for`
 - `break`
 - `continue`
 - `pass`
 - nested loops
-

Module 8: Strings

Everything about Strings

Module 9: Lists

Everything about Lists

Module 10: Tuples

Module 11: Sets

Module 12: Dictionaries

Module 13: Functions

- Built-in
 - User-defined
 - Lambda
 - Recursive
 - Scope
-

Module 14: Exception Handling

- try
 - except
 - else
 - finally
 - raise
-

Module 15: File Handling

- Read

- Write
 - Append
 - CSV
 - JSON
-

Module 16: Modules & Packages

- import
 - from
 - package creation
 - pip
-

Module 17: Object-Oriented Programming (OOP)

- Class
 - Object
 - Constructor
 - Inheritance
 - Polymorphism
 - Encapsulation
 - Abstraction
-

Module 18: Advanced Python

- Iterators
- Generators
- Decorators
- Closures
- Regular Expressions
- Multithreading
- Multiprocessing

- Logging
-

Module 19: Libraries

- NumPy
 - Pandas
 - Matplotlib
 - Seaborn
-

Module 20: SQL with Python

Module 21: Projects

- Calculator
 - ATM
 - Student Management System
 - Library Management
 - Employee Management
 - Chatbot
 - Weather App
 - File Organizer
 - Expense Tracker
-

Module 22: Data Science

- NumPy
 - Pandas
 - Visualization
 - Statistics
-

Module 23: Machine Learning

- Scikit-learn
 - Regression
 - Classification
 - Clustering
-

Module 24: Deep Learning

- TensorFlow
 - PyTorch
-

Module 25: Interview Preparation

- Frequently Asked Questions
- Coding Problems
- MCQs
- Real-Time Scenarios